



Norwegian University of
Science and Technology

FLYT2301: AVIONICS AND FLIGHT SYSTEMS

Lecture 1-2: Aircraft electrical systems and Aircraft Hydraulic Systems

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Overview

Aircraft Electrical Power Generation & Distribution

- Abbreviations

- Overview

- Power Generation

- AC and DC Distribution Architectures

- Power Conversion and Storage

- Smart Power Distribution & Load Management

- Recent Developments: A380 & B787

Aircraft Hydraulic Systems

- Abbreviations

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- Hydraulic Fluid Circuits and Reservoirs

- Engine-Driven, Electric & PTU Pumps

- Accumulators and Actuation Redundancy

- Flight Control and Utility Load Integration

References



Aircraft Electrical Power Generation & Distribution

Aircraft Electrical Power Generation & Distribution

Content

- Moir, I., & Seabridge, A. (2008). *Aircraft Systems: Mechanical, electrical, and avionics subsystems integration* (3rd). John Wiley & Sons, Ltd.
<https://doi.org/10.1002/9780470770931> Chapter 5



Electrical & Power Abbreviations (1/2)

Abbr.	Meaning
AC	Alternating Current
ADP	Air-Driven Pump
APU	Auxiliary Power Unit
ARINC	Aeronautical Radio, Inc. (data-bus standards)
BCRU	Battery Charge/Regulator Unit
BPCU	Bus Power Control Unit
BTB	Bus Tie Breaker
CF	Constant Frequency
CSD	Constant Speed Drive
DC	Direct Current

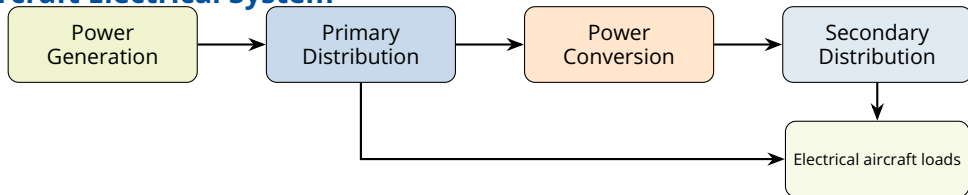
Abbr.	Meaning
EBHA	Electric Backup Hydrostatic Actuator
ECS	Environmental Control System
EDP	Engine-Driven Pump
EE	Electrical Equipment (bay)
EHA	Electro-Hydrostatic Actuator
ELCU	Electronic Load Control Unit
ELMS	Electrical Load Management System
EMP	Electric Motor Pump
EPC	External Power Contactor

Electrical & Power Abbreviations (2/2)

Abbr.	Meaning
GCB	Generator Control Breaker
GCU	Generator Control Unit
IDG	Integrated Drive Generator
IMA	Integrated Modular Avionics
IPP	Integrated Power Package
MTBF	Mean Time Between Failures
PC	Paralleling Contacts
PEPDC	Primary Electrical Power Distribution Centre
PMG	Permanent Magnet Generator
RAT	Ram Air Turbine
RPDU	Remote Power Distribution Unit

Abbr.	Meaning
SCR	Silicon Controlled Rectifier
SEPDB	Secondary Electrical Power Distribution Box
SEPDC	Secondary Electrical Power Distribution Centre
SSPC	Solid State Power Controller
TRU	Transformer Rectifier Unit
VAC	Volts Alternating Current
VDC	Volts Direct Current
VF	Variable Frequency
VFG	Variable Frequency Generation
VSCF	Variable Speed Constant Frequency

The Aircraft Electrical System



Four building blocks

- Power generation – generators driven by engines, APU, or emergency sources
- Primary distribution & protection – switching raw generator power onto main busbars
- Power conversion & energy storage – TRUs, inverters, batteries
- Secondary distribution & protection – delivering conditioned power to individual loads

Electrical Aircraft loads

- motors and actuators generation -
- lighting services
- heating services
- subsystems controllers and avionics systems

Ex: Generic aircraft AC electrical system

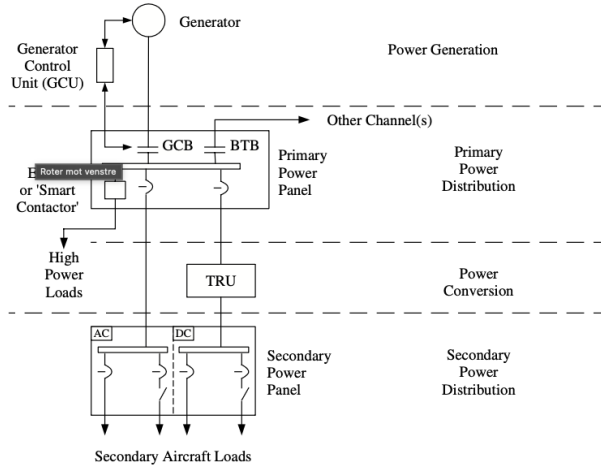


Figure: Moir and Seabridge (2008): Figure 5.3 Generic aircraft AC electrical system

Electrical System Evolution: Electrical Power Generation Growing Fast

From twin 28 VDC to hundreds of kVA

- 1940s–50s: twin 28 VDC generators, one per engine. 5.6–11.2 kVA
- V-bombers: four 115 VAC generators, paralleled for no-break power
- Modern airliners: 20–90 kVA per channel typical
- Airbus A380: 150 kVA per channel
- Boeing 787: 500 kVA per channel – the MEA – more-electric aircraft

Why DC alone does not scale

- DC channels are practically limited to $\sim 400\text{ A}$ ($\sim 12\text{ kW}$ with 28 VDC) by feeder size and contactor switchgear
- AC's higher voltage means lower current for the same power – lower losses, lower weight

Electrical system evolution

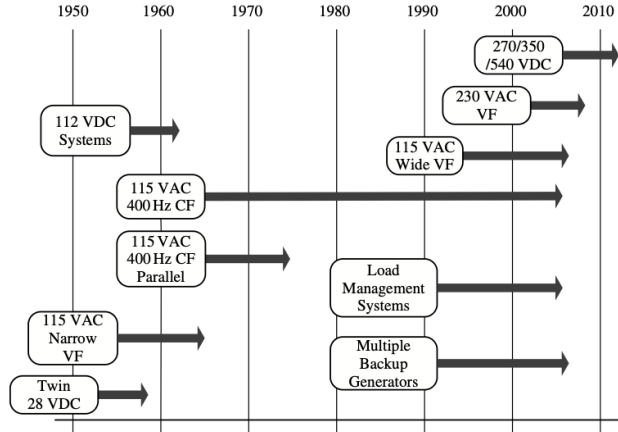


Figure: Moir and Seabridge (2008): Figure 5.2 Electrical system evolution

DC vs AC Power Generation

DC generation

- Self-exciting, **shunt-wound** rotating machine
- A **commutator** rectifies the internal AC to DC
- Output naturally **droops** with load – a voltage regulator trims field current to hold 28 VDC
- Limited to ~ 400 A per channel (conductor size, brush wear)

AC generation

- **Compound generator**: PMG $\rightarrow$ excitation stage $\rightarrow$ main power stage, all on one shaft – no brushes needed for the power stage
- Three-phase, **star-connected**: 115 VAC phase / 200 VAC line, nominally **400 Hz**
- Simpler construction than DC; higher voltage means lower current for the same power

DC Generation: The Shunt-Wound Generator

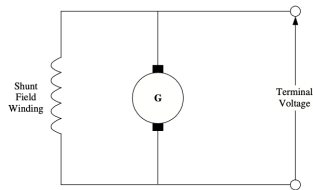
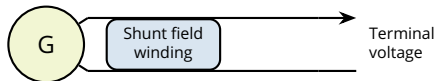


Figure: Moir and Seabridge (2008): Figure 5.4
Shunt-wound DC generator

Self-exciting and self-rectifying

- Rotating electro-magnets generate the power; the **high-resistance field coils** sit in parallel with the armature – the machine excites itself from its own output
- A **commutator** effectively half-wave rectifies and smooths the internal sine-wave output into DC with a residual ripple

DC Voltage Regulation

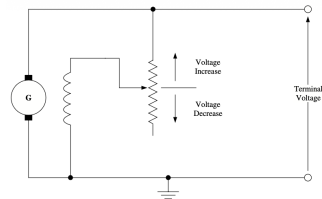
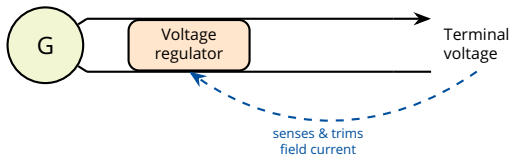


Figure: Moir and Seabridge (2008): Figure 5.5 DC voltage regulator

Countering the natural droop

- Shunt-generator output naturally **droops** with load current; the regulator varies field-circuit resistance to hold a constant **28 VDC** as load and engine speed vary
- Military aircraft regulation must meet **MIL-STD-704D** – the standard for voltage at the point of regulation and permissible drops through the distribution system
- Channel ceiling of **~400 A / 12 kW**: beyond this, conductor/switchgear size and **brush wear** become prohibitive

DC Generators in Parallel: No-Break Power

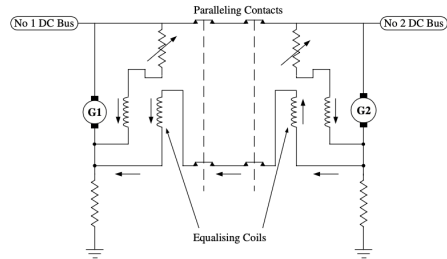
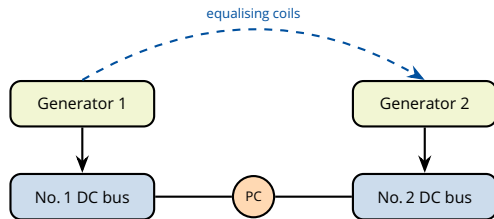


Figure: Moir and Seabridge (2008): Figure 5.8. DC generator parallel operation

Sharing the load equally

- Multi-engine aircraft parallel their generators for **no-break power**: sensitive nav/instrument loads need not restart after a single engine or generator failure
- An **equalising circuit** interlinks the voltage regulators so unequal loading is corrected via each generator's field current
- **PC** = paralleling contacts, closing to tie the two DC buses together

DC System Protection Functions

Fault	Cause	Protective action
Reverse current	Fault allows current to flow <i>into</i> the generator	Reverse-current relay/breaker senses & disconnects the generator
Overvoltage	Field-excitation fault over-excites the generator	Overvoltage sensing opens the line contactor
Undervoltage	Single generator: as reverse current. Paralleled: the equalising circuit tries to raise a lagging generator	Protection is integral to the parallel load-sharing function

Why it matters

Faults must be **contained locally** – a load trip must never propagate to trip the generator itself, and vice versa.

AC generation

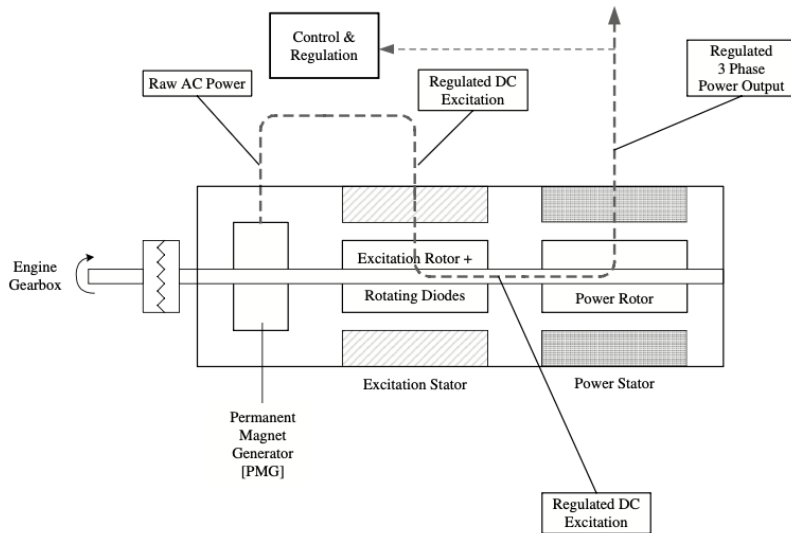
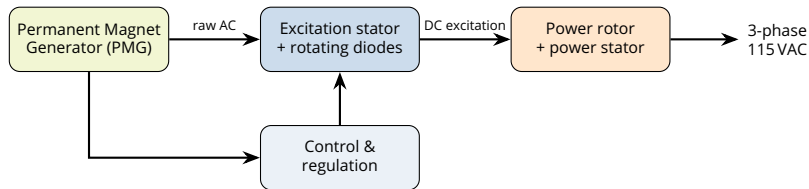


Figure: Moir and Seabridge (2008): Figure 5.6 Principle of operation of modern AC generator

Inside a Modern AC Generator



Three machines on one shaft

The **PMG** bootstraps the system with raw power; the control loop trims **excitation stator** current; the resulting field on the excitation rotor is rectified by **rotating diodes** to excite the main power rotor, inducing regulated 3-phase output in the power stator.

AC Generation: Three-Phase Star Connection

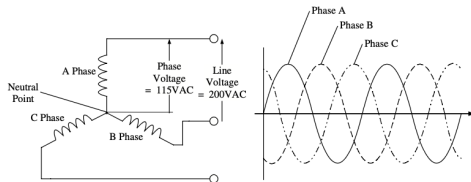
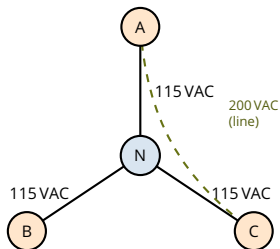
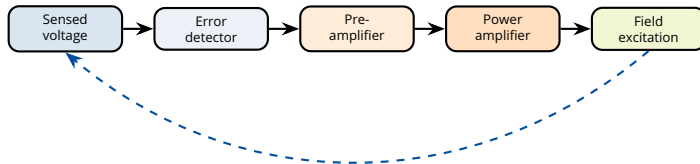


Figure: Moir and Seabridge (2008): Figure 5.7 Star connected 3 phase AC generator

Star-connected three-phase at 400 Hz

- One end of each phase winding is tied to a common **neutral point**; phase voltage (115 VAC) and line voltage (200 VAC) are related by $\sqrt{3}$
- Standard aircraft frequency: **400 Hz**
- Higher voltage → **lower current** for the same power → lower I^2R losses, lower voltage drop, and lighter conductors – AC's main advantage over 28 VDC

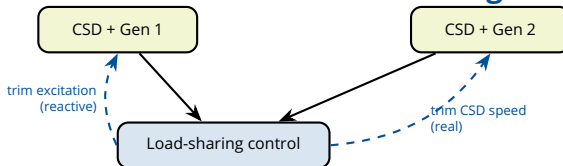
AC Voltage Regulation & Excitation



More complex than DC

- Unlike a self-exciting DC shunt machine, the AC generator needs a **separate DC excitation source** for its field windings – though the PMG allows it to bootstrap itself
- Voltage control uses a closed loop with an **error detector, pre-amplifier and power amplifier** trimming excitation current

AC Generators in Parallel: Real & Reactive Load Sharing



Two independent quantities to balance

- Real load sharing depends on relative rotational speed/phasing – trimmed via the CSD speed
- Reactive load sharing depends on generator output voltage – trimmed via field excitation (using mutual reactors and error-detection circuitry)
- Only applies to constant-frequency generation – variable-frequency machines cannot be paralleled
- Payoff: a single “stiff” vector supply with no changeover transients on a failure, since all generators stay in phase synchronism

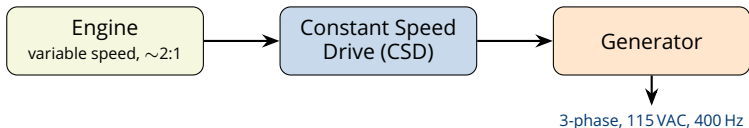
AC Supervisory & Protection Functions

GCU function	What it does
Over/undervoltage	As for DC generation control
Under/over excitation	Detects field current outside its normal range
Under/over frequency	Executed by the real-load-sharing function itself
Differential current	Senses a phase-current imbalance ($> \sim 30$ A) and trips the GCB
Correct phase rotation	Confirms the $R > Y > G$ sequence is not cross-wired

All of this lives in the GCU

The [Generator Control Unit](#) bundles voltage regulation, parallel-operation control and every protection function above into one line-replaceable box per channel.

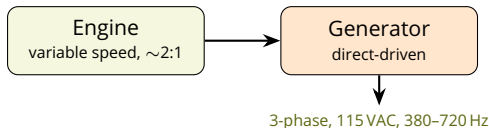
Integrated Drive Generator (IDG)



Most common on today's turbofans

- The CSD acts as a hydro-mechanical automatic gearbox, holding generator shaft speed (and hence frequency) constant to within ~ 10 Hz
- Drawback: the CSD is complex, expensive and needs careful oil maintenance – the least reliable part of the chain

Variable Frequency Generation (VFG)



Simplest and most reliable

- No CSD – the generator is coupled **directly** to the engine gearbox and simply follows engine speed
- Frequency-sensitive loads (AC motors, pumps) may need dedicated **motor controllers**
- Adopted on business jets (Global Express) and now on the **A380** and **B787** (4×150–250 kVA)

Trade-off

VFG cannot be parallelized between channels the way constant-frequency generators can. Separate lines since speed are different

Bridging VF and CF: VSCF

Variable Speed Constant Frequency (VSCF)

Converts frequency-wild generator output into stable 400 Hz **electronically** instead of mechanically – removing the unreliable CSD.

DC link

Raw AC → rectified to an intermediate **DC bus**
→ inverted back to 3-phase 400 Hz. Used on B737, MD-90, B777 (backup).

Cycloconverter

Generates six high-frequency phases (>3000 Hz) and electronically commutates them into 400 Hz output. Proven on F-18, U-2, F-117; not yet adopted in civil service.

Generation Power Levels: A Snapshot

Type	Aircraft	Rating
IDG / CF	B737NG	2 × 90 kVA
IDG / CF	B777	2 × 120 kVA
VF	A380	4 × 150 kVA
VF	B787	4 × 250 kVA
270 VDC	F-22 Raptor	2 × 70 kVA
270 VDC	F-35 Lightning II	1 × 130 kVA (+ 80 kVA IPP)

The trend

More power *and* more diverse generation types on the same airframe – driven by the [MEA](#) concept

The APU as an Electrical Power Source

On the ground

- Provides main aircraft power **without engines running** – for systems checks, cabin services, engine start
- Feeds through an **APU Generator Control Breaker**, controlled by the Bus Power Control Unit (BPCU)

In flight

- Backs up a failed main generator on some aircraft types
- Comparable rating to a main channel – e.g. **90 kVA** on the B767

One of several sources

The primary distribution system accepts power from: main generator (GCB), alternate generator (BTB), **APU generator**, ground power (EPC), and – in extremis – the RAT.

Emergency Power Generation: Ram Air Turbine (RAT)

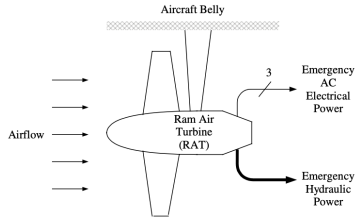


Figure: Moir and Seabridge (2008): Figure 5.22 Ram Air Turbine (RAT)

Deployment

- Extends automatically (or manually) when normal generation is **lost or unavailable**
- Typical sizing: **5–15 kVA** – essential flight instruments and critical services only
- Remains extended for the rest of the flight; restow by **ground maintenance**

Other last-resort options

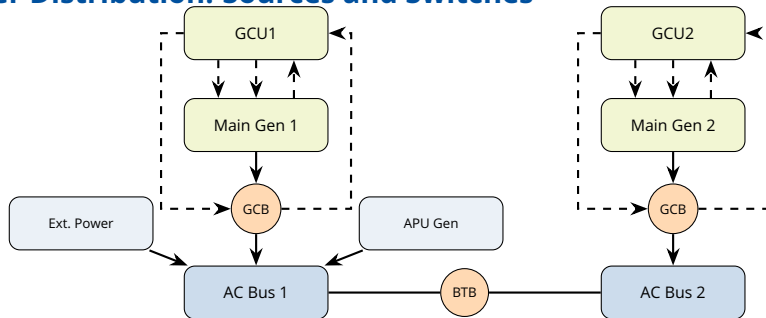
B777 also carries **backup VSCF converters** off the engine gearbox, and **PMGs** feed the flight-control DC system directly for high-integrity, low-power backup.

Primary Power Distribution

Sources

- Main aircraft generator
- Alternate aircraft generator
- Auxiliary power unit [APU](#)
- External/ground power
- Backup converter (B777 only)
- Ram Air Turbine [RAT](#)

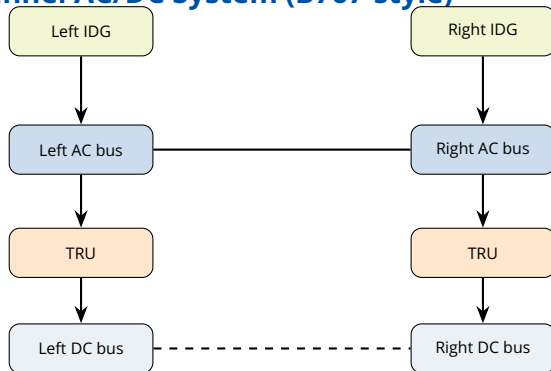
Primary Power Distribution: Sources and Switches



Key switching devices

- **GCB** – Generator Control Breaker, controlled by the **GCU**
- **GCU** – Generator Control Unit, control power, exciter drive (mag. field cont.), sensing (Voltage/Freq./Diff. current))
- **BTB** – Bus Tie Breaker, ties buses together on a source loss
- **APU GCB** and **EPC** (External Power Contactor) – controlled by the **Bus Power Control Unit (BPCU)**
- Cascading rule: a load-level trip must never propagate to trip the next level up

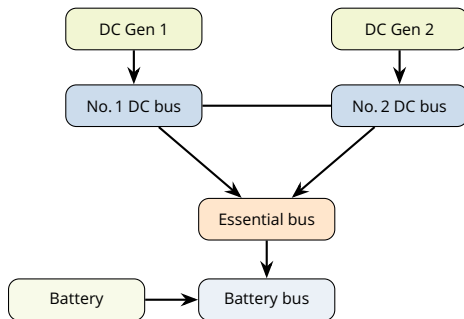
Example: A Twin-Channel AC/DC System (B767-style)



Three independent lanes for autoland

Left bus → left TRU → left DC; right bus → right TRU → right DC; and a third lane through the **battery / hot-battery bus**, fed via a standby inverter – three independent lines of power for the most safety-critical flight phase.

Typical Twin 28VDC System



Suited to smaller aircraft (<~12 kW/channel)

- Two paralleled 28VDC generators feed non-essential loads directly; both feed a **centre/essential bus**
- An inverter off each bus supplies 115 VAC where AC is needed; the battery bus feeds vital services and can back up the essential bus

Power Conversion and Energy Storage

Components

Conversion:

- Inverters
- Transformer Rectifier Units (TRUs)
- Auto transformers
- Battery changers

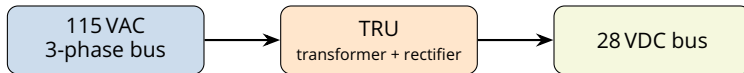
Storage:

- Batteries

Why Convert Power at All? Five recurring conversion needs

- **DC** → **AC**: **inverters** – 28 VDC to 115 VAC
- **AC** → **DC**: **TRUs** – 115 VAC three-phase to 28 VDC
- **AC** → **AC**: auto-transformers – e.g. 115 VAC to 26 VAC lighting
- **Battery charging**: 115 VAC to a regulated 28 VDC charge voltage
- **270 VDC systems** (F-22/F-35): conversion down to 115 VAC/28 VDC for legacy-voltage equipment

Transformer Rectifier Units (TRUs)



The workhorse of AC/DC conversion

- Star primary, dual star/delta secondary windings + 3-phase full-wave rectification
- **B767 example**: 120 A continuous per unit, 180 A for 5 minutes, ~24 lb, forced-air cooled
- Usually **unregulated** – output droops with load; regulated versions exist for demanding applications and can run in parallel

Inverters, Auto-Transformers & Batteries

Inverters

- 28 VDC $\rightarrow$ 115 VAC single-phase
- Feeds essential flight instruments, or backs up **autoland** power
- Modern units: **static**, using fast-switching SCRs to synthesise the AC waveform

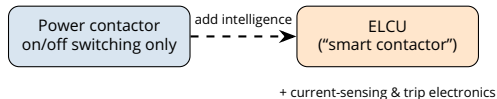
Auto-transformers

Simple step-up/step-down, e.g. 115 VAC $\rightarrow$ 26 VAC for lighting – no isolation, minimal parts

Batteries (Ni-Cd typical)

- Damp transient loads
- Power system **start-up** before generation is online
- Bridge to backup power during an **emergency**
- Capacity in ampere-hours;
 - B767-class aircraft: $\sim$ 30 min standby loads
 - B76 87-class aircraft: $\sim$ 30–60 min depending on battery health, operation length and load mode

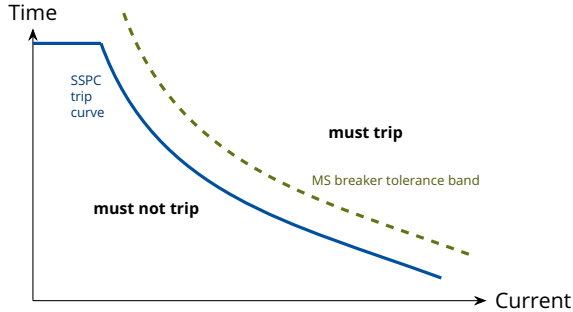
From Contactors to Smart Contactors



Electronic Load Control Units (ELCUs)

- Used for loads above $\sim 20\text{ A}$ per phase ($\sim 7\text{ kVA}$)
- Built-in current-sensing coils on each phase feed a trip-characteristic that can be matched to the load (I^2t , modified I^2t , differential current)
- Report contactor status back to the aircraft system

Circuit Breakers vs Solid-State Power Controllers



Shift from Circuit Breakers to Solid State Power Controllers (SSPC)

- Tighter, electronically-derived trip tolerance than a bi-metallic breaker
- Combine relay/switch and breaker into one solid-state device
- Rated 5/7.5/12.5/22.5 A; costlier today, but more reliable under high duty-cycle switching
- Conventional breakers remain cheap and ubiquitous – 500–600 per aircraft is typical

Electrical Loads

Aircraft Electrical Loads Cat.

- Motors and actuation
- Lighting services
- Heating services
- Subsystem controllers and avionics systems

Table: Aircraft Electrical Load Categories

Category	Primary Role	Safety Priority	Load-Shedding Behavior
Domestic (Hotel)	Passenger comfort and catering	Non-essential	First to be shed
Aircraft Utility	Mechanics, hydraulics, climate	Medium / Phase-dependent	Shed partially or during peak demand
Operations / Avionics	Navigation, mission management	High	Retained unless multiple failures occur
Essential / Flight-Critical	Aircraft control and survival	Critical / Continuous	Never shed (battery/RAT backed)

Case Study: Boeing 777 Electrical Load Management System

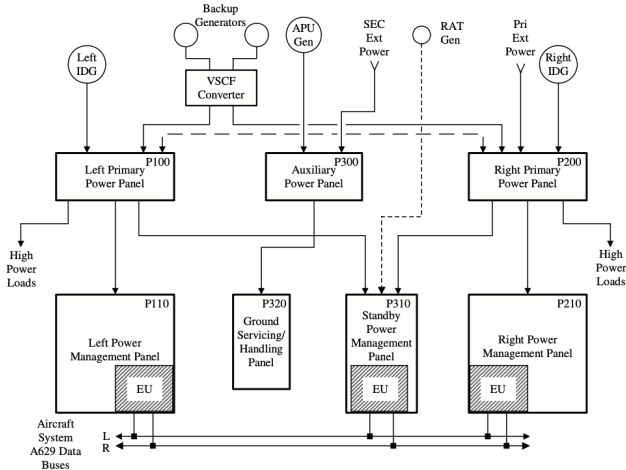


Figure: Moir and Seabridge (2008): Figure 5.26 B777 Electrical Load Management System (ELMS)

First integrated distribution + load-management system (civil)

- Seven power panels; ~18 ELCUs switch loads directly from the main AC buses under embedded intelligence
- Dual-redundant Electronic Units interface via ARINC 629 data buses; modular Line Replaceable Modules (MTBF ~200,000 h)
- Result: ~30% of the on-aircraft test time of contemporary systems, big reductions in wiring, relays and breakers

Load Shed and Optimisation

Beyond simple protection

- On loss of a generation source, ELMS **automatically reconfigures** loads for the optimum use of remaining power – not just an emergency shed
- When power is restored, loads are **re-instated on a schedule**, not all at once
- Embedded utility functions: fuel jettison, **automatic RAT deployment**, refuel/defuel control, fire suppression, APU start

The shift in philosophy

Electrical distribution moves from **passive protection** (breakers, contactors) to **active management** of power as a shared, prioritised resource across the aircraft.

Airbus A380: Power Generation

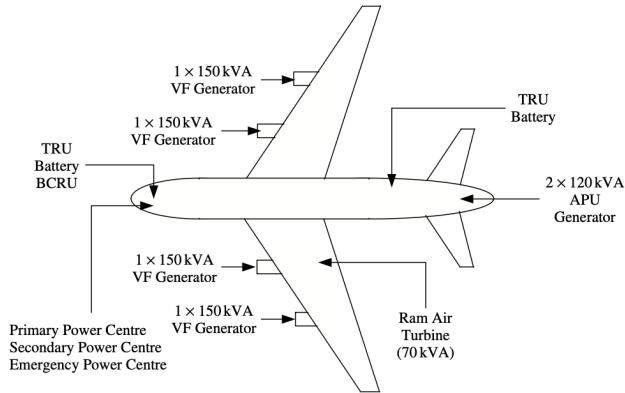
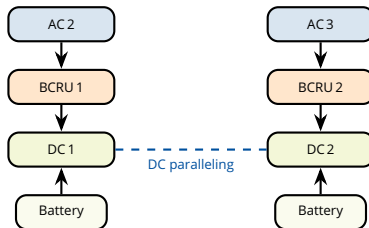


Figure: Moir and Seabridge (2008): Figure 5.37 A380 – power system components

The first large civil re-adopter of VF

- Four 150 kVA VF generators (370–770 Hz), one per engine – a record for civil aircraft when introduced (B777 offered $120 \text{ kVA CF} + 20 \text{ kVA VSCF}$ backup, $\sim 140 \text{ kVA/channel}$)
- Backed by $2 \times 120 \text{ kVA CF APU generators}$, a 70 kVA RAT , and four external ground-power connections
- Because to VF: the four main channels **cannot be paralleled** – frequency given by each engine's speed independently

A380: Engineering No-Break Power on the DC Side



A clever workaround

- Since the VF AC channels can't be paralleled, no-break power is engineered on the DC side instead
- BCRUs (regulated TRUs) convert AC to a paralleled 28 VDC network feeding the IMA cabinets and control computers – power that must never interrupt
- Three $\times$ 300 A BCRUs/TRU + three $\times$ 50 Ah batteries give three independent, continuously-available DC channels

A380: Segregation and Power Distribution

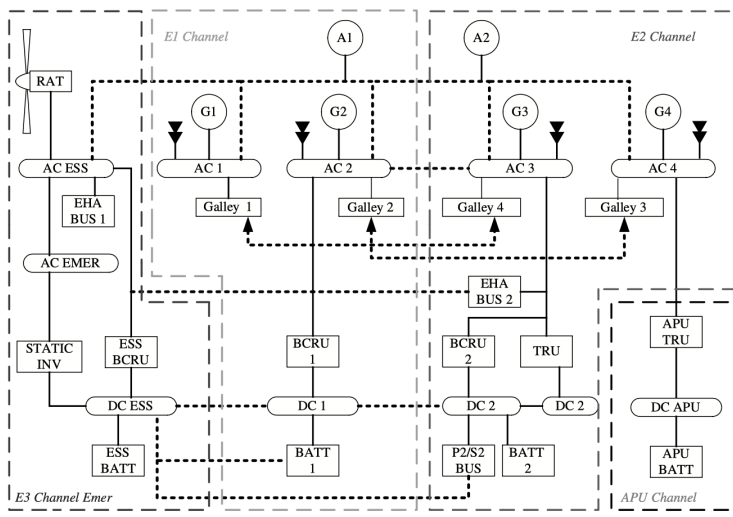


Figure: Moir and Seabridge (2008): Figure 5.40 A380 total electrical system showing segregation

A380: Segregation and Power Distribution

Four independent electrical channels

- Channel E1: powered by AC Gen. 1 & 2
- Channel E2: powered by AC Gen. 3 & 4
- Channel E3: powered by RAT & static inverter
- Channel APU: APU start
- E1, E2, E3: an associated main generator, a BCRU and a 50 Ah battery for an independent power path
- Distribution hierarchy: PEPDC (primary) → SEPDCs (aircraft loads <5 kVA) → SEPDBs (domestic loads, dispersed near the cabin to save feeder weight)
- SSPCs are preferred over thermal circuit breakers throughout secondary distribution

Domestic load	Typical power
Cabin lighting	~15 kVA
Galleys (intermittent, meal service)	120–240 kVA
Galley cooling (continuous)	~90 kVA
In-flight entertainment (continuous)	~50–60 kVA (~100 W/seat)

Boeing 787: The “Bleedless” Engine

Function	Conventional (bleed air)	B787 (electric)
ECS & pressurisation	Engine bleed air	4× electric compressors (~500 kVA)
Wing anti-ice	Hot bleed air	Electric heating (~100 kVA)
Engine start	High-pressure air cart (Air Start Unit / Ground Cart) / APU / engine	Starter/generators (~180 kVA)
Hydraulic pumps	EDP + bleed-driven ADP	4× electric motor pumps (~400 kVA)

Only one bleed left

The **sole** remaining bleed-air use is low-pressure fan air for **engine cowl anti-icing** – everything else moves to electricity, motivated by the disproportionate penalty of bleeding air from increasingly high-bypass, high-pressure-ratio engines.

B787: Power Generation & Distribution

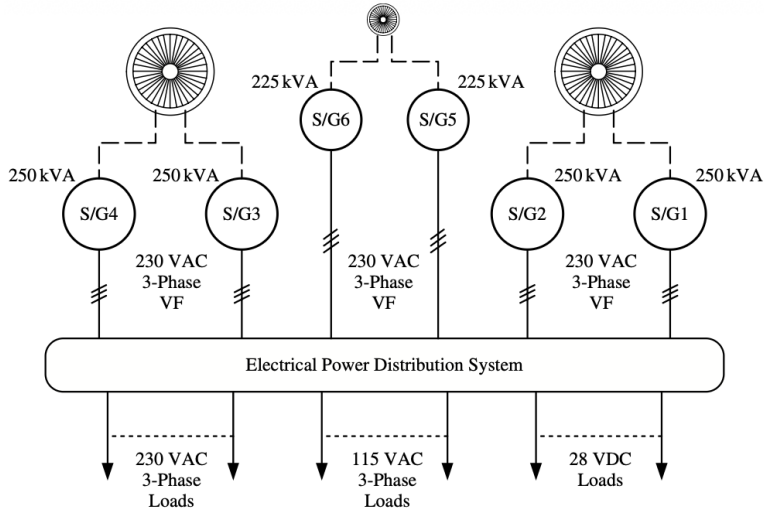
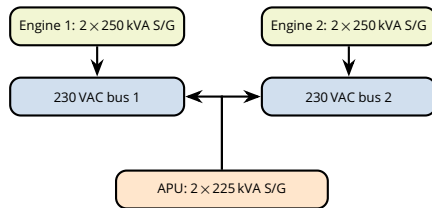


Figure: Moir and Seabridge (2008): Figure 5.41 Boeing 787 top-level electrical system

B787: Power Generation & Distribution



500 kVA per channel at double the voltage

- 230 VAC three-phase (400 VAC line-to-line) instead of the industry-standard 115 VAC – halves current for the same power, cutting feeder losses and wiring weight, at the cost of extra care against corona discharge
- Power covered to 115 VAC/28 VDC for legacy-voltage subsystems
- Distribution: 4 liquid-cooled primary panels (fwd/aft EE bays) plus 21 Remote Power Distribution Units (RPDUs) for secondary loads

B787: Electric Air Conditioning

Stage	Conventional (bleed, pre cooled)	Electric (B787)
From source	$\sim 200^{\circ}\text{C}$, 30 – 45 psi	$\sim 100^{\circ}\text{C}$, 15 psi
Delivered to cabin	$\sim 15 - 18^{\circ}\text{C}$, 11.8 psi	$\sim 15 - 18^{\circ}\text{C}$, 11.8 psi

Four motor-compressors – two packs

- Two air-conditioning packs, each with **two electrically-driven compressors** (~ 125 kVA per permanent-magnet motor)
- Output passes through ram-air-cooled heat exchangers, then mixes with recirculation air – same delivered condition as bleed air, without tapping the engine core
- Electric ECS starts from **much lower temperature and pressure** – the gap versus bleed air is wasted energy in the conventional approach

A380 vs B787: Two Philosophies

Aspect	A380	B787
Primary power	115 VAC VF, 4×150 kVA	230 VAC VF, 2×500 kVA
Bleed air	Retained conventionally	Nearly eliminated ("bleedless")
No-break power	Engineered on the DC side (BCRU paralleling)	Multiple starter/generator channels + battery
Hydraulics	2H + 2E: two conventional circuits + EHA/EBHA	Conventional 3-channel (L/C/R), 5000 psi, 4×EMPs

Same pressure – different answers

Relative B777 generation: Both aircraft push per-channel power up by roughly 3–4x, but the A380 stays closer to convention while the B787 commits fully to the more-electric concept.

Summary: Electrical Power Generation & Distribution

What we covered

- **Generation:** DC vs AC principles; IDG (CSD) vs VFG vs VSCF; the APU and RAT as backup sources
- **Distribution:** primary AC/DC bus architectures, GCB/BTB/APU-GCB/EPC switching, twin-DC and three-lane autoland examples
- **Conversion:** TRUs, inverters, auto-transformers, batteries & chargers
- **Smart distribution:** ELCUs, SSPCs, and the B777 ELMS as the state of the art in load management
- **Recent developments:** the A380 and B787 as two design choices to address higher power demand

Next

Hydraulic Systems – the other major power-transmission medium on the aircraft, and its own story of redundancy and load integration.



Aircraft Hydraulic Systems

Aircraft Hydraulic Systems

Content

- Moir, I., & Seabridge, A. (2008). *Aircraft Systems: Mechanical, electrical, and avionics subsystems integration* (3rd). John Wiley & Sons, Ltd.
<https://doi.org/10.1002/9780470770931> Chapter 4



Hydraulic & Power Abbreviations

Abbr.	Meaning
ADP	Air-Driven Pump
AMAD	Aircraft Mounted Accessory Drive
APU	Auxiliary Power Unit
DC	Direct Current
ECAM	Electronic Crew Alerting & Monitoring (Airbus)
EDP	Engine-Driven Pump
EICAS	Engine Indication & Crew Alerting System (Boeing)
FC	Flight Control(s)
FCS	Flight Control System

Abbr.	Meaning
IDG	Integrated Drive Generator
MIL-H	US fluid specification (Military Handbook)
N2	Engine core (HP-spool) rotational speed
NRV	Non-Return Valve
PTU	Power Transfer Unit
RAT	Ram Air Turbine
SOV	Shut-Off Valve
SV	Servo-Valve

Why Hydraulics?

A power-transmission workhorse

- Appeared in the early 1930s with the retractable undercarriage
- Purpose: Transfers small, low-energy cockpit inputs into high-energy action at control surfaces or gear(s)
- Powers primary & secondary flight controls, undercarriage, brakes, anti-skid, and many utility functions

Why not all-electric?

- Low weight per unit power – the electric motor still cannot match a hydraulic actuator's power/weight ratio above $\sim 3 \text{ kW}$
- Small, flexible pipework; oil provides self-lubrication
- Actuators can stall or overload without damage and recover normally

Design drivers

- Low weight, low volume
- Low initial cost
- High reliability
- Low maintenance

Safety-critical

- Flight-critical hydraulic functions must not fail totally under a single failure.
- Redundant systems: multiple pumps, accumulators and leak isolation.

Hydraulic Fluid Circuits

Hydraulic Functions

- Primary flight controls
 - Elevators (1)
 - Rudders (2)
 - Ailerons (3)
 - (Flaperons (4))
 - Canards ("front wing")
- Secondary flight controls
 - Flaps (5)
 - Slats (6)
 - Spoilers (8)
 - Airbrakes (9)
 - Stabilizer trim (10)
- Utility systems
 - Landing gear
 - Brakes
 - Steering gear
 - Aerial refueling probes (military)
 - Cargo door
 - Loading ramp (military)
 - Passenger stairs

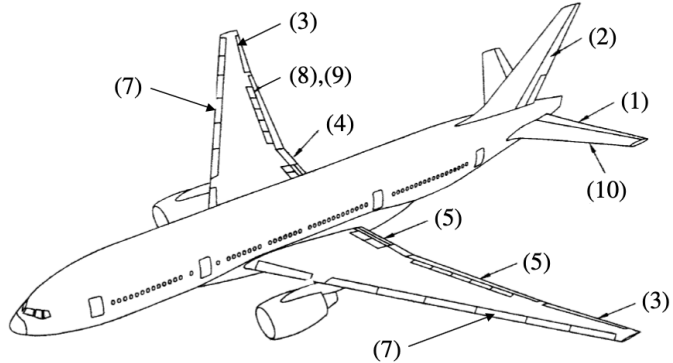


Figure: Moir and Seabridge (2008): Figure 4.1 Hydraulic system loads

A Hydraulic System

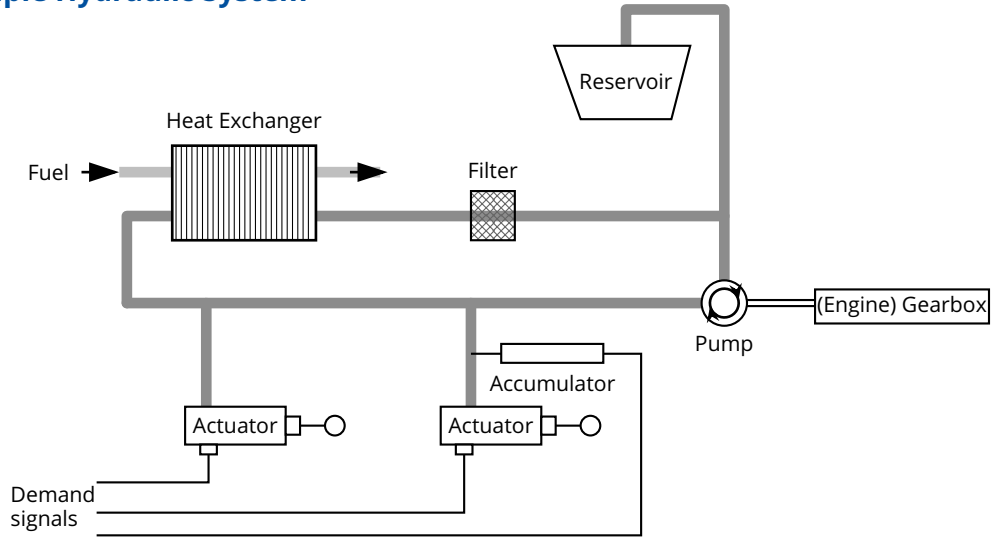
Core components

- an **energy** source (engine, APU or RAT)
- a **reservoir**
- a **filter** (clean hydraulic fluid)
- a **distribution** network of pipes, valves and shut-off cocks
- pressure and temperature **sensors**
- a mechanism for **cooling** hydraulic oil
- **demand** delivery; **actuators**, **motors**, **pumps** and
- an **accumulator** for storing **energy**

Safety

- A single circuit cannot meet flight-safety requirements: Recurring use of **multiple, segregated circuits**

Simple Hydraulic System



Hydraulic Fluid: Type, Pressure, Temperature

Fluid types

- **Military:** mineral-based fluid – MIL-H-5606 (US) / DTD 585 (UK) – cheap, available, but flammable and limited to $\sim 130^{\circ}\text{C}$
- **Civil:** phosphate-ester fluids (e.g. Skydrol) – fire-resistant but not fireproof
- Synthetic fluids (MIL-H-83282) raise the temperature limit to $\sim 200^{\circ}\text{C}$

Standard working pressures

- Industry standard: 3000 psi or 4000 psi
- Studies of 5000–8000 psi promise lower mass/volume, but DTD 585 shears above 5000 psi
- Optimum pressure differs per aircraft – standardisation wins for cost and shared components

Flow rate is a system-level trade

Peak demand (e.g. several actuators moving together) sets pipe and pump sizing; short, high-velocity transients are often better met by an **accumulator** than by oversizing the pump.

The Reservoir and Fluid Conditioning

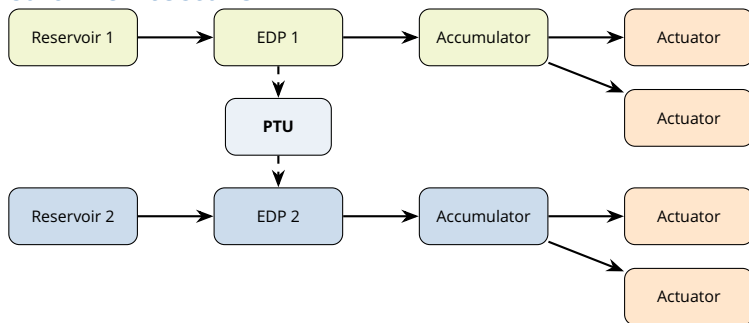
Reservoir

- Pressurised (typically by regulated **bleed air**) to ensure good pump volumetric efficiency
- Military reservoirs must be **fully aerobatic**: no air/fluid interface, fluid supply maintained at all attitudes and g
- Sized for differential system volumes, thermal expansion and an emergency margin
- Level switches allow **isolation of leaking sections** to protect flight-critical circuits

Cooling & cleanliness

- A fuel/hydraulic **heat exchanger** cools the return line before fluid re-enters the reservoir
- Fine servo-valve clearances demand very clean fluid: typically a **5 μm absolute** return-line filter with a **15 μm** pressure-line filter
- Electronic particle counters verify cleanliness levels for ground crews

Redundant Circuit Architecture



Segregation

- Independent circuits are routed **geographically apart** to avoid common-mode damage (accidental or battle damage)
- Military aircraft: typically **two** circuits. Large civil transports: **three or more**
- A **Power Transfer Unit (PTU)** lets one circuit hydraulically power another with *no fluid transfer* between them

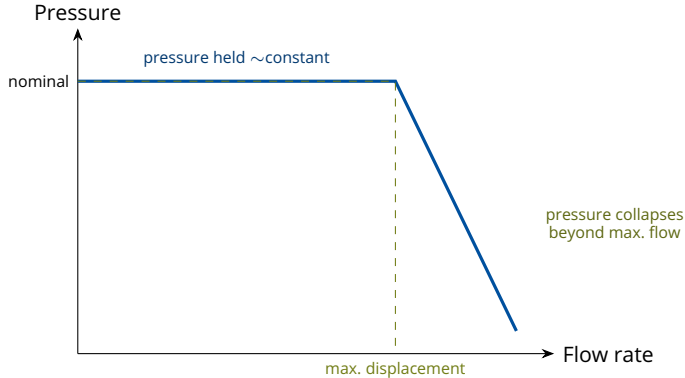
Hydraulic Pump Technologies

Pump type	Displacement	Max. pressure	Notes
Gear	Fixed	<1500 psi	Contamination tolerant
Gerotor	Fixed	<1000 psi	Gearbox lube systems
Vane	Fixed/variable	~2000 psi	Catastrophic failure modes
Piston	Fixed/variable	>5000 psi	Standard for aircraft hydraulics
Centrifugal	–	<100 psi	Very high reliability

The variable-displacement piston pump

- Nine pistons reciprocate against a **swash plate**; plate angle sets stroke and hence flow
- **Variable delivery, constant pressure**: the pump self-adjusts displacement to hold pressure within ~5% of nominal as demand varies
- At maximum plate angle, further flow demand causes pressure to **collapse rapidly** – the accumulator bridges short peak demands

Pump Characteristic: Constant-Pressure Behaviour



Bridging peak demand

An **accumulator** (compressed-gas cylinder) augments flow beyond pump capability for short periods; the pump must then have time to recharge it before repeated use empties it.

Pump Power: EDP, Electric & Air-Driven Pumps

Engine-Driven Pump (EDP)

- Mounted on the engine accessory gearbox (civil) or an **Aircraft Mounted Accessory Drive** (military)
- Primary, continuously-rated power source; speed follows engine rpm

Electric motor-driven pump

- AC or DC motor driven; **backup** or **ground-servicing** source
- Automatically latches on if EDP delivery pressure drops below a threshold

Power Transfer Unit (PTU)

- Back-to-back hydraulic **motor + pump**: one system's pressure drives a pump feeding another system
- No fluid crosses between circuits – preserves segregation
- Activates automatically on a pressure-differential trigger

Air-Driven Pump (ADP)

- Bleed-air turbine driven; used as a demand/backup pump (e.g. B767 centre system)

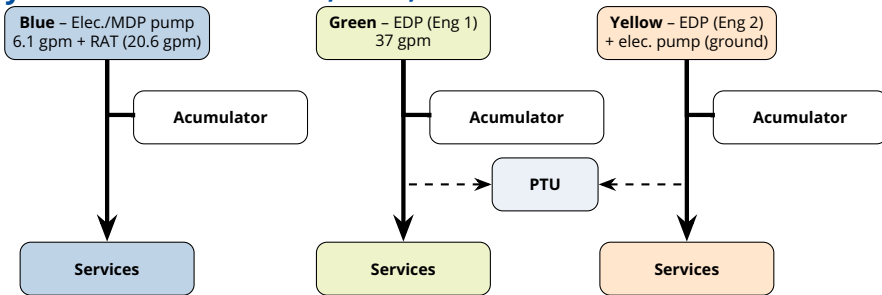
Hydraulic Fluid/Circuits Encoding

Manufacturer/OEM Specific

Different aircraft manufacturers designate redundant hydraulic circuits using specific color names or letters:

- Airbus, BAE Systems (and Concorde): Named exclusively by color — Green, Blue, and Yellow. Cockpit controls, electronic centralized aircraft monitor (ECAM) pages, and system schematics are directly color-coded around these names
- Boeing: Uses letter designations — typically System A, System B, and Standby (on narrow-bodies like the 737) or Left (L), Center (C), and Right (R) (on wide-bodies like the 777/787).
- Other Manufacturers: Many regional and military aircraft (e.g., Bombardier, Dassault, Lockheed) use numbers (System 1, 2, 3).

Case Study: Airbus A320 – Green, Blue, & Yellow Circuits



Three continuously-operating systems

- Each system has its own reservoir; **leak-measurement** and **priority valves** protect flight-critical loads under low pressure
- Bi-directional **PTU** links Green and Yellow (triggers above 500 psi differential)
- **RAT** auto-extends on total engine + APU loss, feeding the Blue system

Case Study: Airbus A320 – Green, Blue, & Yellow Circuits

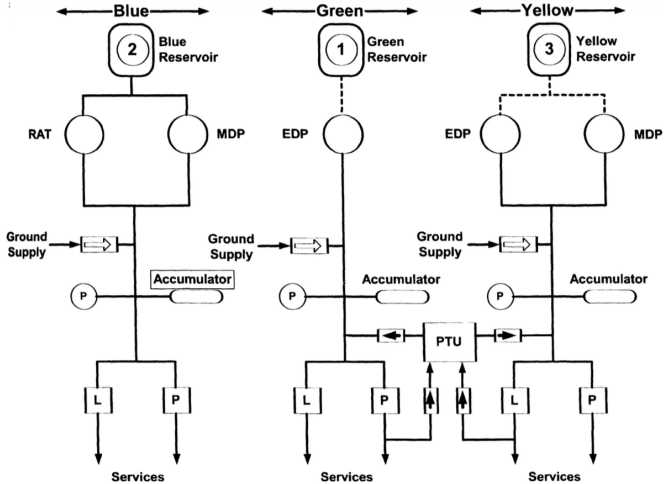


Figure: Moir and Seabridge (2008): Figure 4.20 Simplified A320 family hydraulic system

Example: A380 Break Control System – Green Circuit(s)

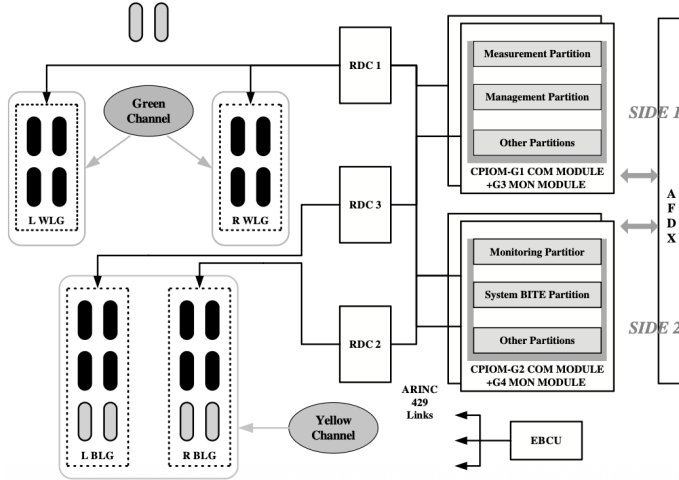


Figure: Moir and Seabridge (2008): Figure 4.31 A380 brake control system

Case Study: Boeing 767 – Left, Right, Centre

System	Continuous pump	Demand / backup pump
Left	EDP (37.5 gpm)	Electric motor pump
Right	EDP (37.5 gpm)	Electric motor pump
Centre	2× electric motor pumps	ADP + RAT (emergency)

Eight pumps – one philosophy

Boeing favours [more pumps with diverse prime movers](#) over extensive accumulator use. The [RAT](#) auto-deploys once both engines fall below 50% N2 with airspeed above 80 kt, and can only be restowed on the ground.

Case Study: Boeing 767 – Left, Right, Centre

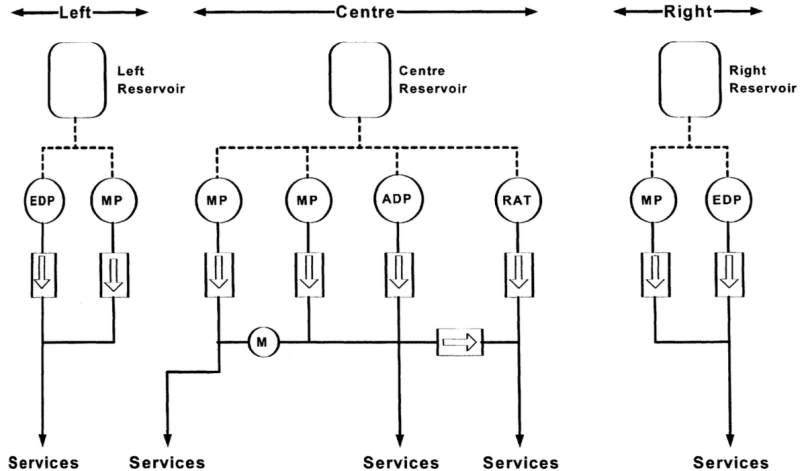


Figure: Moir and Seabridge (2008): Figure 4.21 Simplified B767 hydraulic system

The Role of the Accumulator

Stored hydraulic energy

- A compressed-gas cylinder that augments pump flow for **short, high-velocity** transients
- Stabilises pressure during PTU and standby-generator run-up
- Fluid volume and pressure available depend on **gas temperature**

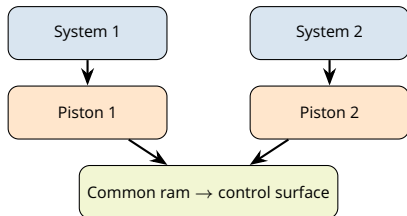
Mandatory emergency roles

- Standby **wheel-brake** accumulator – guaranteed brake applications with all pumps lost
- Emergency **canopy** opening
- Enough flight-control authority to recover to level flight for crew ejection

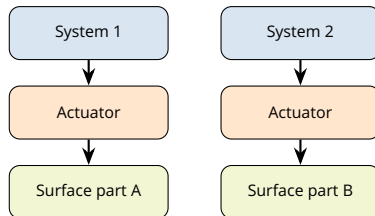
The accumulator only helps if it is **recharged between demands** – the pump must have spare capacity and time, or repeated use will empty it.

Redundant Actuation: Military vs Civil Philosophy

Military: tandem actuator



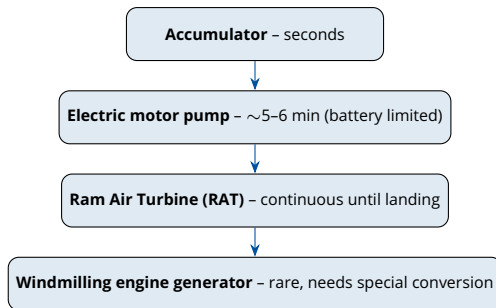
Civil: split control surface



Two ways to survive a single failure

- **Military:** two pistons in tandem on one ram, each fed by a different circuit, acting on one control point
- **Civil:** each control surface is **split** into independent parts, each with its own actuator and hydraulic system

Emergency Power Sources: A Hierarchy



Increasing duration – decreasing power

Each layer trades **available power** for **duration**: an accumulator delivers a burst; a RAT can sustain flight-critical services for the remainder of the flight, but only ~5–15 kVA.

Classifying Hydraulic Loads

Primary flight controls

- elevators, rudders, ailerons, flaperons, canards
- **Flight-safety critical**: no single failure may prevent or interrupt operation

Secondary flight controls

- flaps, slats, spoilers, airbrakes, stabilizer trim
- Performance may **degrade** in a controlled and announced way

Utilities / services

- Gear, brakes, steering, doors, stairs, refuelling probe
- May be **expendable** after failure, or one-shot emergency selection

Special case: Wheel-brakes

Wheel-brakes are a special case: power is drawn from up to **three sources**, including a stored-energy accumulator that also serves the parking brake.

Sizing a Hydraulic System

Six questions the designer must answer

- **Pressure** – what nominal system pressure (3000/4000 psi)?
- **Integrity** – flight-safety critical, or can loss be tolerated? How many independent sources?
- **Flow rate** – angular/linear rate or volume/second to achieve the action
- **Duty cycle** – continuous (e.g. primary FC on an unstable airframe) vs occasional (gear, twice per flight)
- **Emergency / reversionary use** – does this circuit back up another system (e.g. a hydraulically driven electrical generator)?
- **Heat load** – energy dissipated into the fluid by each component

Integration Example: Avro RJ Yellow & Green

Yellow system	Green system
Flap motor, roll spoilers, inner lift spoilers, rudder servo, standby fuel pumps, gear emergency lock-down, wheel brakes & park brake, airstairs	Flap motor, centre/outer lift spoilers, airbrakes, normal gear, nose-wheel steering, wheel brakes (excl. park)

Hydraulics powering electrics

The Green system can drive a standby AC/DC generator through a hydraulic motor – a direct link between the two systems covered in this lecture: hydraulic power backing up the electrical system when needed.

Lecture 1-2: Electrical & Hydraulic Foundations

- **Electrical:** power generation (IDG, VFG, APU, RAT), AC/DC distribution, power conversion (TRUs, inverters), and smart load management (ELCUs, SSPCs, ELMS)
- **Hydraulic:** fluid circuits & reservoirs, engine-driven/electric/PTU pumps, accumulators & redundant actuation, and flight-control/utility load integration

The common thread

Both type of systems are built from the same design pattern:

- redundant sources
- segregated distribution
- stored/emergency energy, and
- prioritised loads

References

References I



Moir, I., & Seabridge, A. (2008). *Aircraft Systems: Mechanical, electrical, and avionics subsystems integration* (3rd). John Wiley & Sons, Ltd. <https://doi.org/10.1002/9780470770931>

